

Economics 220:322:01 - Econometrics

Syllabus Fall 2025

This syllabus is subject to change throughout the semester. Announcements in class, via e-mail, or on Canvas supersede anything written here.

Instructor: Hector Blanco

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New Jersey Hall, Room 205A

Office Hours: Tuesdays 11am-12pm in-person (or e-mail for appointment via Zoom or in-person). Use this Zoom [link](#).

Lectures: Tuesdays and Thursdays 3.50pm to 5.10pm at Frelinghuysen Hall (General Purpose Classroom): FH-A6

Teaching Assistant: Shaojun Tong

E-mail: st1113@economics.rutgers.edu (put Econ 322 in the subject!)

Office Hours: Wednesdays 10-11am via Zoom (or e-mail for appointment via Zoom). Use this Zoom [link](#).

Final exam: Thursday, December 18th, 4pm to 7pm

Last day to drop without a W: September 11th

Last day to withdraw with a W: October 27th

Any withdrawal after October 27th will result in an “F” grade

Course Description: In this course, you will start the process of becoming a consumer and producer of empirical economic research. You will learn to apply statistical methods to estimate and test economic relationships. The methods we learn in this course have a wide range of applications in many different areas—many firms now employ economists who are trained to use econometric tools in a variety of settings!

Course Objectives and Learning Goals: Students who successfully complete Econ 322 should be comfortable with basic statistics and probability. They should be able to use a statistical/econometric computer package to estimate an econometric model and be able to report the results of their work in a non-technical and literate manner. In particular, a student who successfully completes Econ 322 will be able to estimate and interpret linear regression models and be able to distinguish between economic and statistical importance. They should be able to critique reported regression results in applied academic papers and interpret the results for someone who is not trained as an economist.

Prerequisite(s): It will be expected that all students have passed the following courses with a C or better: principles of economics courses covering both microeconomics and macroeconomics (220:120, Introduction to Microeconomics; and 220:103, Introduction to Macroeconomics); Math 135 (640:135, Calculus I); and an introductory statistics class (960:211 or 960:285 Introduction to Statistics, or equivalent). I will assume that you have good command of the material covered in these classes, particularly Statistics. While there will be an overview of the required statistical knowledge at the beginning of class, I strongly suggest that you review the material from your introduction to statistics course if it has been a few semesters since you took this class.

Important Note: Students majoring in Economics need to pass this course with a C or higher. If by the middle of the semester you fear you will not reach this goal please contact me ASAP—do not wait until the end of the semester to try to remedy a poor performance as it will be too late!

Textbook: James Stock and Mark Watson Introduction to Econometrics, 4th Edition (Any edition is really fine, *in particular feel free to buy used copies*. There is also an eTextbook option in the Pearson website.)

Software: This class will use R. R is a data and statistical analysis software increasingly used in Economics. If you are already familiar with another statistical software package (like Julia, Python, SAS, SPSS, or Stata) and wish to use that for your problem sets, you are welcome to. However, no tutorials or code hints will be given for other languages.

R is free to use and download. We will have an in-class tutorial on using R and resources will be posted on Canvas throughout the semester with hints and tricks.

Communication Policy:

- Ask questions related to class materials during lectures.
- Use the online forum Piazza. We will have threaded discussions (monitored by the instructor and the TA) for class-related topics, e.g., lectures, problem sets, and exams. I encourage you to use (and often check out) this resource: you will not only be able to get your questions answered but you will also benefit from seeing the answers to other students' questions. This resource should allow you to get a timely, high-quality answer to most of your questions. You can sign up using this [link](#).
- Ask questions during the instructor's office hours and TA office hours (see above).
- For personal class-related matters, e-mail is the best option: hector.blanco@rutgers.edu. All e-mails regarding this class must have "Econ 322" in the subject—this helps me filter important e-mails, so I can make sure to respond to you.

Communication about Assignments and Readings:

Announcements will be made via Canvas announcements. All students are required to check their Rutgers e-mail and the Canvas website—I would recommend logging in at least once a day. The way you have notifications set up on Canvas will determine whether important announcements get e-mailed to you or not.

Course Evaluation:

- **Problem Sets (30%):** There will be five problem sets. These assignments will provide hands-on practice and application of the statistical and econometric concepts covered in the course using R (or the software of your choice).
- **Empirical Project Proposal (10%):** Towards the end of the semester you will be required to turn in a proposal for an empirical project that would use the knowledge you learned in this class to answer an interesting question. A proposal will consist of: an answerable question of economic interest, a description of the data you would need in order to answer this question, a description of the econometric technique you would use to answer the question, and your hypothesis about the results based on economic theory. You do not actually have to complete the proposed project. Project proposals should be at maximum 3 pages in length, double spaced in 12pt Times New Roman font. The due date for the Project Proposals is: **Thursday, December 4th, 8pm**. More instructions will be posted on Canvas and announced in class later in the semester.
- **Midterm Exams (20%):** Two midterm exams will be held during class time. Midterms are cumulative, though more importance will be given to new material—and you often need to know earlier material in order to understand and perform later material. There will be two midterms—only your highest score on the two midterms will count towards your final grade. Midterm dates will be announced in class and via Canvas at least 1 week before they happen. Note that because lowest scores are dropped, make-ups will not be provided except in extenuating circumstances discussed with the professor.
- **Final Exam (40%):** The final exam is on **Thursday, December 18th, 4pm to 7pm**. The final exam will be cumulative and include all announced readings, textbook chapters, and lecture notes.

Course Policies

- **Academic Dishonesty:** Please review the University's Academic Integrity policy: <http://academicintegrity.rutgers.edu/>. Academic misconduct will be taken very seriously. Any assignment for which you are explicitly allowed to work in groups should still be written up in your own words and turned in individually unless otherwise specified. You may not take any unauthorized materials into an exam.
- **Makeup exams** are only provided for extenuating circumstances. Barring extenuating circumstances, you must notify me within 24 hours of the missed exam.

- If you have a final exam conflict (more than 2 final exams on one calendar day, more than 2 final exams in consecutive periods, or 2 exams scheduled for the same time) please let me know as soon as you know and a make-up will be scheduled if required by the policies of the School of Arts and Sciences.
- **Regrading Policy:** If you believe your score on a mid-term or assignment was penalized incorrectly, you must turn in the exam or assignment with a typed description of why you want a particular question regraded and a clear and complete explanation of why you believe you deserve more credit. These should be handed back to me in class or in office hours within a week of receiving the exam or assignment. Obviously, if there is a simple addition error please bring it to our attention right away and we will fix it.
- **Late Work Policy:** Late work (Problem Sets or the Empirical Project Proposal) will be penalized 10% per 24 hours late, after 3 days (from 24 to 72 hours late) any remaining late work will be penalized 50%; no late work will be accepted past that point. I recognize that sometimes ‘life happens’—as such you have two allotted ‘flex days’. These days allow you to submit an assignment up to two days late without penalty (assignments are Problem Sets or the Empirical Project Proposal the). You can use these for Problem Sets or the Empirical Project Proposal. You do not need to provide me with a reason, simply fill out [this form](#) specifying the assignment for which you will use the flex days and how many flex days you want to use (*if you do not do this and turn in something late, I will assume you are not using flex days*). You must fill the form within 48 hours of the problem set deadline.

Course Outline:

This is a rough guide to topics we will discuss and textbook chapters that go along with each topic in parenthesis. Additional readings will be announced via Canvas and in class.

1. Introduction to Econometrics and Causality (Chapter 1)
 - Reading: Introduction Mastering 'Metrics by Angrist & Pischke, available for download here: <http://press.princeton.edu/chapters/i10363.pdf>
2. Review of Concepts in Probability and Statistics (Chapters 2 and 3)
3. Univariate Linear Regression (Chapters 4 and 5)

Note: Midterm I will happen around here.
4. Multivariate Linear Regression (Chapters 6 and 7)
5. Non-linear regression (Chapter 8)
 - Non-linear Regression
 - Polynomials

- Logarithms
 - Interactions
6. Assessing Regression Models (Chapter 9)
- Internal Validity
 - External Validity
- Note: Midterm II will happen around here.
7. Panel Data Models (Chapter 10)
8. Binary Dependent Variables (Chapter 11)
9. Additional Topics (time-permitting, readings will be announced):
- Instrumental Variables
 - Quasi-experimental Methods: Differences-in-Differences, Regression Discontinuity
 - Time-Series

Student-Wellness Services:

- **Counseling, ADAP & Psychiatric Services (CAPS):** <http://health.rutgers.edu/medical-counseling-services/counseling/>. (848) 932-7884. 17 Senior Street, New Brunswick, NJ 08901. CAPS is a University mental health support service that includes counseling, alcohol and other drug assistance, and psychiatric services staffed by a team of professionals within Rutgers Health services to support students's efforts to succeed at Rutgers University. CAPS offers a variety of services that include: individual therapy, group therapy and workshops, crisis intervention, referral to specialists in the community, and consultation and collaboration with campus partners.
- **Crisis Intervention:** <http://health.rutgers.edu/medical-counseling-services/counseling/crisis-intervention/>
- **Report a Concern:** <http://health.rutgers.edu/do-something-to-help/>
- **Violence Prevention & Victim Assistance (VPVA):** <https://vpva.rutgers.edu/>.(848) 932-1181. 3 Bartlett Street, New Brunswick, NJ 08901. The Office for Violence Prevention and Victim Assistance provides confidential crisis intervention, counseling and advocacy for victims of sexual and relationship violence and stalking to students, staff and faculty. To reach staff during office hours when the university is open or to reach an advocate after hours, call 848-932-1181.

- **Disability Services:** <https://ods.rutgers.edu/>. (848) 445-6800. Lucy Stone Hall, Suite A145, Livingston Campus, 54 Joyce Kilmer Avenue, Piscataway, NJ 08854. Rutgers University welcomes students with disabilities into all of the University's educational programs. In order to receive consideration for reasonable accommodations, a student with a disability must contact the appropriate disability services office at the campus where you are officially enrolled, participate in an intake interview, and provide documentation: <https://ods.rutgers.edu/students/documentation-guidelines/>. If the documentation supports your request for reasonable accommodations, your campus's disability services office will provide you with a Letter of Accommodations. Please share this letter with your instructors and discuss the accommodations with them as early in your courses as possible. To begin this process, please complete the Registration form on the ODS web site at: <https://ods.rutgers.edu/students/registration-form/>.